



Experimental and numerical investigation of turbulent flow and heat transfer characteristics in the Komax static mixer

Huibo Meng, Tong Meng, Yanfang Yu*, Zongyong Wang, Jianhua Wu

Engineering and Technology Research Center of Liaoning Province for Chemical Static-mixing Reaction, School of Mechanical and Power Engineering, Shenyang University of Chemical Technology, Shenyang 110142, Liaoning, China

ARTICLE INFO

Article history:

Received 1 February 2022

Revised 1 April 2022

Accepted 7 May 2022

Available online 16 May 2022

Keywords:

Komax static mixer

Turbulent flow

Heat transfer

Field synergy

Nusselt number

Power spectrum

ABSTRACT

The turbulent flow and heat transfer characteristics in the Komax static mixer are experimentally and numerically investigated in the range of Reynolds number (Re) from 2640 to 17600. The vortex evolution, Nusselt number (Nu), Darcy friction factor (f), PEC , field synergy principle and power spectrum in the Komax are evaluated based on that in Ross LPD, respectively. The Computational Fluid Dynamics (CFD) predictions of Nu and f based on SST $k-\omega$ model from ANSYS Fluent have a good agreement with the experimental results. The magnitudes of longitudinal vortices and swirl intensity in Komax are larger than those of Ross LPD. The synergy performance between velocity and temperature fields in Komax is better than that in Ross LPD due to the Dean Vortex gradually induced and weakened by the Komax inserts. The Komax static mixer has an obvious superiority in heat transfer and could enhance Nu higher than Ross by 46.7%–49.6% at the cost of friction factor increasing 2.80–4.37 times. Considering the PEC of both static mixers greater than 1, the Komax insert are recommended for $Re < 5280$ and Ross LPD would be a good alternative due to much lower pressure drop for $Re \geq 5280$. Based on the uniform heat flux experiments, the empirical correlations of Nu and f in both static mixers are established with the prediction errors less than $\pm 6.20\%$. Turbulent kinetic energy and turbulent dissipation rate are used to evaluate the mesoscopic and microscopic mixing performance. The power spectrum and scaling relationship of outlet pressure pulsation signals are evaluated to reveal the turbulent chaotic performance which could help to enhance the mass and heat transfer. The scale relationship between power spectrum and pressure pulsation frequency satisfies power function attenuation.

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1. Introduction

For significant reducing of mixing time and energy consumption, it is often necessary to strengthen the heat and mass transfer process. The purpose of heat transfer enhancement technology is using less energy to transfer more heat [1]. The convective heat transfer enhancement in the tube is of great significance in the continuous process intensification industries. It is well known that heat transfer enhancement technology includes active and passive enhanced heat transfer method. Passive heat transfer is widely used in the tube heat exchangers due to low cost and ease of installation [2–4]. Bejan and Kraus [5] further categorized the flow pattern in the tube into two parts: core flow and boundary flow. For the traditional convection heat transfer enhancement technology, more attention is paid on the boundary layer where has a

large temperature gradient. Therefore, previous research on heat transfer enhancement technology mainly focused on the effective disturbance on the boundary layer. In addition, Liu and Yang [6] argued that the core flow could be fully utilized to enhance heat transfer. Under the condition of the boundary layer disturbed as much as possible, heat transfer enhancement in the core flow area can be further achieved. The traditional passive enhanced heat transfer technology was consisted of enhanced heat transfer surfaces and properly selecting working fluid [7]. Compared with enhanced heat transfer surfaces including modified surfaces or geometries, tube inserts had been widely concerned because of low energy consumption and high performance [8]. Among the inserts, the extensive attention had been attracted on twisted tape and static mixer, and the understanding to the twisted tape is more comprehensive [9–11].

The heat transfer enhancement theory about twisted tape was earlier studied by Smithberg and Landis [12], who experimentally studied the influence of twist ratio and working fluid for heat transfer enhancement and empirical equations of Nusselt number

* Corresponding author.

E-mail address: taroyy@163.com (Y. Yu).

Nomenclature

A_r	aspect ratio
B	boundary layer number
$c_{p,w}$	specific heat at constant pressure of water, J/(kg·K)
$c_{p,s}$	specific heat at constant pressure of stainless steel, J/(kg·K)
D	diameter of the pipe, mm
f	friction factor
f_0	friction factor of a smooth tube
h_{conv}	convective heat transfer coefficient, W/(m ² ·K)
L	length of the pipe, mm
l_1	inlet section, mm
l_2	length of the mixing element, mm
l_3	distance between adjacent elements, mm
l_4	outlet section, mm
Nu	Nusselt number
Nu_0	Nusselt number of a smooth tube
P	power consumption of static mixer, W
P_w	power spectrum of outlet pressure pulsation, dB
Pr	Prandtl number
PEC	Performance evaluation criterion
Δp	pressure drop, Pa
Q_v	volume flow, m ³ /s
q	heat flux, W/m ²
Re	Reynolds number
Se	secondary flow dimensionless number
S	swirl number
T_{in}	mean temperature of the inlet, K
T_{out}	mean temperature of the outlet, K
ΔT_m	logarithmic mean temperature difference, K
u_m	mean axial flow velocity, m/s
x, y, z	three dimensional coordinates, m
Greek symbols	
α	angle of twist, °
β	volume-weighted average synergy angle, °
δ	thickness of element, mm
ε	turbulent dissipation rate, m ² /s ³
ϕ	scaling exponent of power spectrum
η	thermal performance factor
θ	volume-weighted average synergy angle, °
k	turbulent kinetic energy, m ² /s ²
λ_s	thermal conductivity of stainless steel, W/(m·K)
λ_w	thermal conductivity of water, W/(m·K)
μ_w	hydrodynamic viscosity of water, kg/(m·s)
ρ_s	density of stainless steel, kg/m ³
ρ_w	density of water, kg/m ³

(Nu) and friction factor (f). And for fully developed turbulent flow in tubes with twisted tape, the velocity field is helicoidal and corresponds to a forced vortex in the core superposed on an essentially uniform axial flow. Manglik and Bergles [13,14] experimentally evaluated the heat transfer characteristics of water and ethylene glycol using three twisted tapes with different twist ratios and the correlated equations of Nu and f for laminar and turbulent flows were combined. They concluded that twisted tape provided a long flow path and secondary flow for heat transfer enhancement. Murugesan et al. [15] experimentally investigated Nu , f and thermal performance factor (η) in a circular tube fitted with full-length twisted tape with trapezoidal cut in the range of Reynolds number (Re) between 2000 and 12000. Compared with those in the plain tube, the trapezoidal-cut twisted tape is reasonable for the conspicuous improvement in the heat transfer coefficient and friction factor. Farhad et al. [16] numerically investigated the Nu , f

and η in the cases of using twisted tap or perforated twisted tap in converging-diverging tubes. They concluded that perforated holes in twisted tap had an inapparent effect on heat transfer due to a considerable reduction in f up to 396% and improvement in η up to 56% compared to smooth diverging tubes. Bhattacharyya et al. [17] conducted a numerical study of heat transfer characteristics in a tube with an inserted twisted tape vortices generator. The results showed that a higher entrance angle yielded a higher η and the heat transfer and pressure drop can be greatly improved by using single twisted tape compared with the traditional one. Wang et al. [18] experimentally and numerically investigated heat transfer characteristics of polygonal twisted elements in the constant wall temperature with Re ranging from 200 to 1800. The results revealed that the entransy efficiency increased with increasing edge of polygonal twisted elements. Smith Eiamsa-ard and their colleagues [19–21] numerically analyzed the temperature and velocity fields in the circular tube with twisted tape and studied the enhanced heat transfer mechanism. It was found that short-length twisted tapes generated strong helicoidal flow at the entrance and the vortices intensity and heat transfer rate gradually reduced after through the twisted tape. Full-length twisted tapes had better use-value compared with short one. However, the heat transfer enhancement and flow characteristics of twisted tape with alternate clockwise and counterclockwise had more superiority than those of the full-length twisted tape in the laminar flow.

Static mixers as a high-efficiency mixing unit operation are increasingly used in continual mixing and heat transfer process industries [22]. The research on the mixing performance of standard Kenics static mixers (KSM) had been relatively comprehensive [23,24]. With the increasingly severe situation of international environmental protection, the research of static mixers as inserts in multifunction heat exchanger has also received more and more attentions on efficiently utilizing and controlling the heat generated or needed in the production processes [25]. Joshi et al. [26] experimentally investigated the heat transfer coefficient of KSM and obtained the empirical correlation between f and Nu under $Re=0.1-50000$ and aspect ratio $A_r=1.5-2.5$. Meng et al. [27] numerically investigated forced convective heat transfer of water in the modified Kenics static mixer (MKSM) in steady turbulent flow. Thermal performance factor decreased gradually with increasing Re which indicated that the MKSM segments were more effective for heat transfer in lower Re . Habchi et al. [28,29] experimentally and numerically investigated the effect of elements arrangements in High-efficiency vortex (HEV) static mixers on fluctuating velocity, f , Nu , η and other parameters. It could be found that the mixing and heat transfer performance in the alternating array HEV static mixer increased significantly compared with that in the aligned array ones. Genetti [30] experimentally investigated heat transfer of KSM and Ross LPD static mixer in the laminar flow and established the correlation between the Nu and f in the range of $Re=10-1000$. The results showed that the Ross LPD static mixer gave rise to more heat transfer enhancement than that in the KSM. Meng et al. [31] used numerical methods to study the turbulent flow and heat transfer features in the Lightning static mixer (LSM) in the range of $Re=4000-30000$ and evaluated the enhancement mechanism of flow field by the vortex's evolution, field synergy principle and entropy generation analysis. Yu et al. [32] numerically investigated the laminar characteristics of heat transfer and flow friction in tube equipped with two Q types of RL- and RR-series inserts. Based on the numerical results, synergies between velocity and temperature fields had been investigated. Because of the effects of the flow division, flow reverse and radial flow, node-weighted average synergy angle is the smallest at the transition of mixing elements. Kwon et al. [33] modified the structure of KSM by adding a new wing. The results demonstrated the potential of additive manufacturing to enable novel flow geometries can enhance convection

heat transfer whilst minimizing pressure drop penalties and volumes. Konopacki et al. [34] experimentally and numerically studied and analyzed flow field, temperature distribution and heat transfer efficiency in a novel static mixer. The result indicated that the novel static mixer may be used as a heat exchanger in the laminar flow.

Tu et al. [35–37] proposed a new kind of inserts and systematically studied the heat transfer performance and flow resistance of the inserts in laminar and turbulent conditions. They found that the inserts could promote the cold fluid shift from the core to the wall with little effect on flow, which could achieve high heat transfer efficiency and low resistance loss. Liu et al. [38] experimentally and numerically investigated the turbulent flow and heat transfer characteristics in a circular tube fitted with rectangular winglet vortices generators. Based on numerical data, new correlations of Nu and f were also developed using four non-dimensional parameters of winglet heights, slant angle, Prandtl number (Pr), and Re .

Until now, the Komax static mixers had been successfully used in distributing the water and bottom sediments across the crude oil pipeline for accurate crude oil sampling and crude oil metering. Nevertheless, inadequate work has been reported about the turbulent flow and heat transfer performance of a tube with the Komax static mixer [39]. Consequently, the turbulent flow resistance and heat transfer performance in the Komax static mixer are experimentally and numerically investigated in this paper. In particular, this paper aims at three major issues of enhancement performance of Komax static mixer: (i) the effects of Komax static mixer and Ross LPD static mixer on heat transfer, flow friction and overall thermal performance are studied within the range of Re from 2640 to 17600, (ii) reliable empirical correlations of Nu and f based on the experimental data are obtained, (iii) the superiority of Komax static mixer in mesoscopic and microscopic mixing performance are highlighted by comparison with that of Ross LPD static mixer.

2. Experimental apparatus and procedure

2.1. Geometry of the elements

The Komax static mixer (Komax Systems, Inc.) achieves high efficiency through triple action mixing. They are designed for use where additives to the main pipeline flow have already been introduced upstream of the mixer. The Komax mixing elements are more applicable in the granular flow compared to Ross (Ross Engineering, Inc.). However, the use of Ross is more financially acceptable due to its simpler geometry and lower pressure drop [40]. The turbulent flow, heat transfer and mixing characteristics in the Komax are analyzed based on those in the Ross LPD. The two different configurations of static mixers studied are shown in Fig. 1(a). The design parameters of Komax and Ross LPD elements has been described as shown in Fig. 1(b) and 1(c). The Ross LPD mixer consists of a series of semi-elliptical plates which are discriminately positioned in a smooth tube. A single element consists of two plates perpendicular to each other as shown on Fig. 1(c). The specific parameters of the geometries are given in Table 1.

2.2. Experiment set up and Measurement techniques

The heat transfer and flow resistance experimental apparatus included a stainless-steel tube heat exchanger, motionless mixer inserts, a manometer, thermocouples, pump, feed tank, temperature and pressure receivers as shown in Fig. 2. The experimental test tube is consisted of three sections: inlet section (1000 mm), test section (850 mm) and outlet section (1500 mm). Komax and Ross LPD elements are fixed on a 2 mm wide square steel and installed on the test section, respectively. Both models

of inserts made of polylactic acid (PLA) materials could be obtained by a 3D printer (HORI-Z600, Beijing Huitianwei Technology Co., Ltd.). Through XH-W3001 thermal control module and customized thermal insulation heating sleeve with constant thermal power, the following relationship between wall heat flux and Re is guaranteed:

$$q = 2.273Re + 7000 \quad (1)$$

The thickness of 304 stainless steel tube used in the experiment is 1.55 mm. Six K-type contact thermocouples with an accuracy of $\pm 0.5\%$ are selected to measure the wall temperature and high-efficiency heat-conducting silicone grease is smeared between the wall and the thermocouples. Ignoring the heat loss induced by conduction, the average values obtained from six thermocouples can be used as the final wall temperature. Two plug-in PT 100 temperature sensors are used to measure the fluid temperature at the inlet and outlet, respectively. The calibration accuracy of the temperature sensors is $\pm 0.2\%$. The centrifugal pump (MH1802, Wilo) with the maximum flow rate of $12 \text{ m}^3/\text{h}$ and uncertainty of $\pm 4\%$ is used to pump the cold fluid into the experimental pipe. The flow of water is measured by stainless steel rotameter (LZD-50/Y10/RR1/ESK, Shenyang Beixing Instrument Co., Ltd.) with 1.5 accuracy class and glass rotor flowmeter (VA10-25, Changzhou Xunyi Instrument Co., Ltd.) with 4 accuracy class in the range of $1\text{--}10 \text{ m}^3/\text{h}$ and $0.16\text{--}1.6 \text{ m}^3/\text{h}$, respectively. The inlet and outlet pressure are measured by pressure converter (SIN-PX300, Hangzhou Liance Instrument Co., Ltd.) with the range of $0\text{--}50 \text{ kPa}$ and the accuracy of $\pm 0.5\%$. The backflow volume of the hot fluid is adjusted to ensure that the inlet temperature of cold-water tank is 293K and the error is not more than 2% . The on-line high speed data acquisition DEWE-3021 (Dewetron GmbH, Austria) and multi-channel temperature data loggers AT4716V (Changzhou Applent Instruments Co., Ltd.) are used for the measurement of the pressure and temperature, respectively.

2.3. Data processing and uncertainty calculation

The experiments for turbulent flow resistance and heat transfer were conducted, respectively. Water is used as the working fluid and the range of inlet flow rate is from $0.3 \text{ m}^3/\text{h}$ to $2 \text{ m}^3/\text{h}$. The corresponding Re ranges from 2640 to 17600. Parameters used in the experiment and data processing are as follows:

$$Re = \frac{\rho_w u_m D}{\mu_w} \quad (2)$$

where u_m is the average inlet velocity, D is the pipe diameter, μ_w is the hydrodynamic viscosity of water, and ρ_w is the density of water.

The Prandtl number (Pr):

$$Pr = \frac{c_{p,w} \mu_w}{\lambda_w} \quad (3)$$

where $c_{p,w}$ is specific heat at constant pressure of water; λ_w is the thermal conductivity of water.

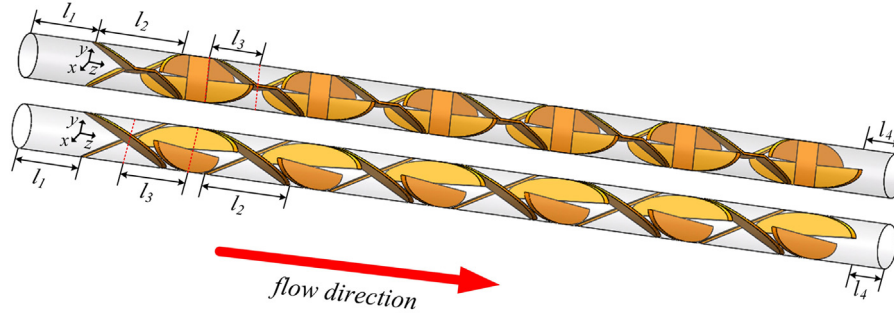
The heat transfer coefficient (h):

$$h = \frac{q}{\Delta T_m} \quad (4)$$

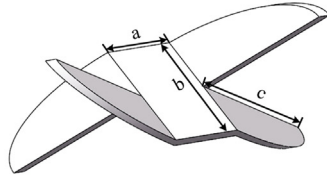
The average Nusselt number (Nu):

$$Nu = \frac{hD}{\lambda_w} \quad (5)$$

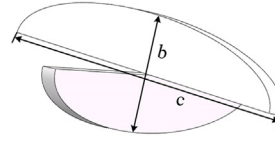
where ΔT_m is the logarithmic mean temperature difference.



(a) Geometries of the tube fitted with different static mixer



(b) Komax static mixer



(c) Ross LPD static mixer

Fig. 1. Geometries of Komax and Ross LPD static mixer in different views.**Table 1**
Specifications of the Komax & Ross LPD static mixer.

Specification	Dimension (Komax)	Dimension (Ross LPD)
Entrance section (l_1)	40mm	40mm
Length of the mixing element (l_2)	92.75mm	82.91mm
Distance between adjacent elements (l_3)	40mm	60mm
Exit section (l_4)	60mm	60mm
a	20mm	-
b	40mm	40mm
c	40mm	88.11mm
Diameter of the pipe (D)	40mm	40mm
Angle of twist (α)	30°	30°
Number of elements	12	12
Thickness of element (δ)	2mm	2mm

The relationship between Darcy friction factor (f) and pressure drop is:

$$f = \frac{2D\Delta p}{\rho_w L u_m^2} \quad (6)$$

where Δp is the pressure drop of the measured section.

Uncertainties in Re , Nu and f can be calculated from:

$$\frac{U_{Re}}{Re} = \left[\left(\frac{U_D}{D} \right)^2 + \left(\frac{U_{\mu_w}}{\mu_w} \right)^2 + \left(\frac{U_{\rho_w}}{\rho_w} \right)^2 + \left(\frac{U_{w_m}}{w_m} \right)^2 \right]^{0.5} \quad (7)$$

$$\frac{U_{Nu}}{Nu} = \left[\left(\frac{U_D}{D} \right)^2 + \left(\frac{U_{\lambda_w}}{\lambda_w} \right)^2 + \left(\frac{U_{h_{conv}}}{h_{conv}} \right)^2 \right]^{0.5} \quad (8)$$

$$\frac{U_f}{f} = \left[\left(\frac{U_D}{D} \right)^2 + \left(\frac{U_L}{L} \right)^2 + \left(\frac{U_{\rho_w}}{\rho_w} \right)^2 + \left(\frac{U_{w_m}}{w_m} \right)^2 + \left(\frac{U_{(\Delta p)}}{(\Delta p)} \right)^2 \right]^{0.5} \quad (9)$$

Based on Table 2 and Eqs. (7)–(9), the maximum uncertainties in Re , Nu and f in this work are 4.01%, 6.01% and 4.04%, respectively.

Table 2
Uncertainty of each parameter in the experimental system.

Parameters	Absolute uncertainty	Relative uncertainty
Pipe dimeter (D)	± 0.1 mm	
Pipe length (L)	± 1 mm	
Temperature (T)		$\pm 0.2\%$
Pressure (p)		$\pm 0.5\%$
Heat flux (q)		$\pm 6\%$
Volume flow (Q_v)		$\pm 4\%$

2.4. Verification of experimental reliability

In order to verify the accuracy of the experimental results, the experimental data in the smooth tube are compared with the empirical correlation. The experimental correlations proposed by Gnielinski and Blasius, Filonenko, Petukhov and Dittus-Boelter in the smooth tube are shown as follows [41,42]:

Blasius:

$$f = 0.3164 Re^{-0.25} \quad (10)$$

Filonenko:

$$f = (1.82 \lg Re - 1.64)^{-2} \quad (11)$$

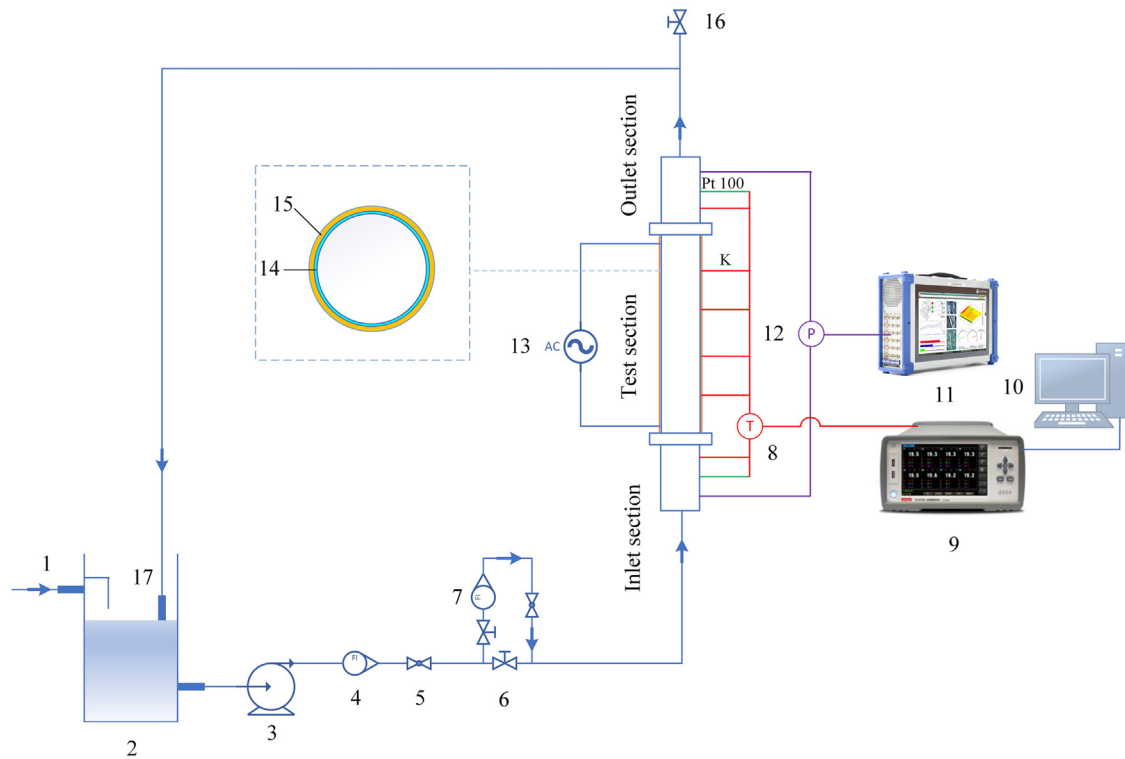


Fig. 2. Schematic diagram and testing apparatus of experiment.

1. Cold water inlet 2. Cold water tank 3. Wilo centrifugal pump 4. stainless steel rotameter 5. Globe valve 6. Check valve 7. glass rotor flowmeter 8. Thermocouple 9. multi-channel temperature data loggers 10. Computer 11. DEWE-3021 12. Pressure sensors 13. Alternating current power supply 14. 304 stainless steel tube 15. Heating unit 16. Air release valve 17. Drainage exit.

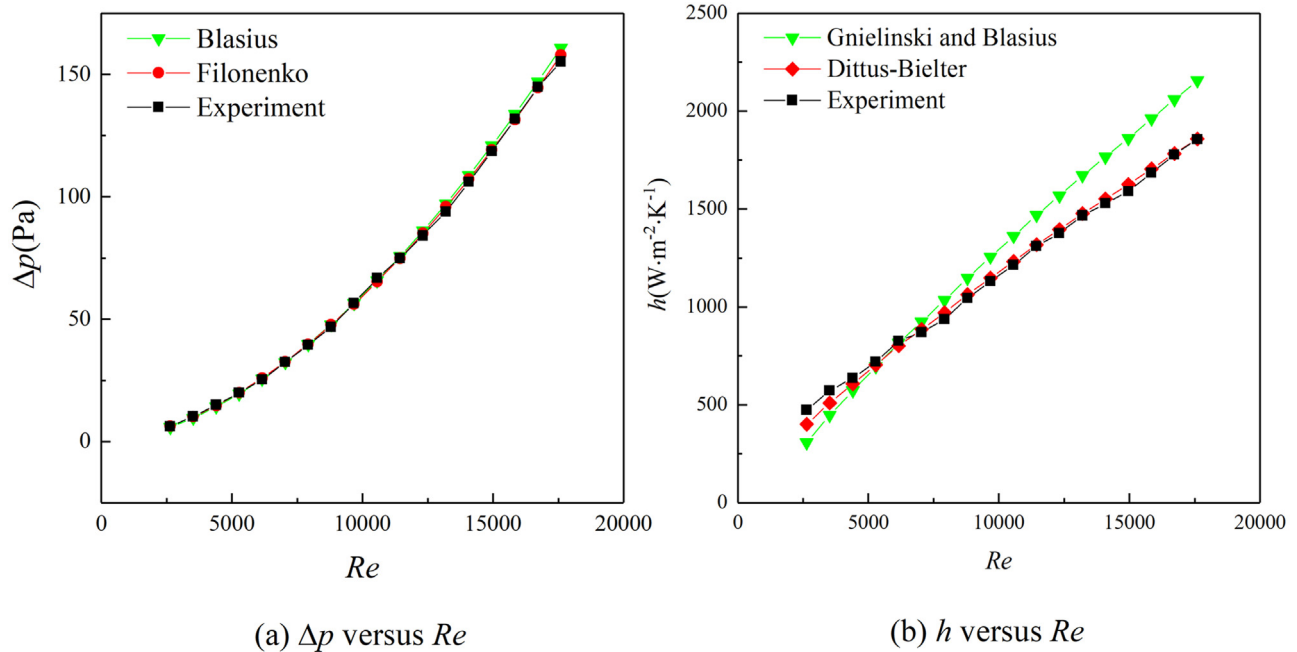


Fig. 3. Validations of the h and Δp of smooth tube.

Gnielinski and Blasius:

$$Nu = \frac{(f/8)(Re - 1000)Pr_f}{1 + 12.7\sqrt{f/8}(Pr_f^{2/3} - 1)} \left[1 + \left(\frac{d}{l} \right)^{2/3} \right] c_t \quad (12)$$

where $c_t = (Pr_f/Pr_{wall})^{0.11}$, $Pr_f/Pr_{wall} = 0.05 \sim 20$.

Dittus-Bielter:

$$Nu = 0.023Re^{0.8}Pr^{0.4}c_t \quad (13)$$

where $c_t = (\mu_f/\mu_{wall})^{0.11}$

Fig. 3 shows the validation tests about h and Δp in the smooth tube between the experiment values and empirical correlations. Compared with the correlation values from Filonenko and Blasius,

Table 3
Initialization parameter for grid independence verification of Komax.

Boundary layer number (<i>B</i>)	0	5	7	10	13	15
Grid number of Komax	1286541	1789682	1983482	2259021	2537017	2703221

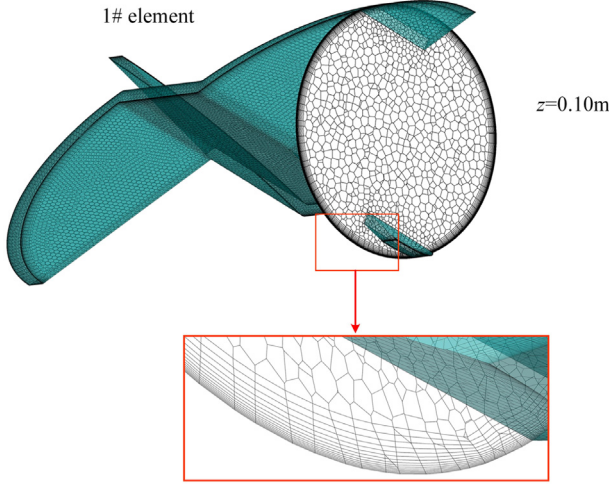


Fig. 4. The grid on the cross-sectional plane at $z=0.10$ m of Komax static mixer.

the maximum deviations of Δp as shown in Fig. 3(a) are 2.58% and 5.99%, respectively. In Fig. 3(b), the experimental h is similar to those obtained from Dittus-Boelter and Gnielinski correlations. The maximum deviations of heat transfer coefficient are 11.04% and 22.03%, and the proportions of deviation values below 10% are 83.3% and 94.4%, respectively. It is concluded that the experimental measurements have good agreements with the empirical correlations, which proves that the experimental system is reliable.

3. Numerical simulation

3.1. Grid independence test

For the complicated flow field, polyhedral mesh is used to discretize the whole computational domain. The number of grids is less compared with those of other types under the condition of ensuring the calculation accuracy. Fig. 4 shows the cross-section mesh at $z=0.10$ m of the first element in Komax static mixer. It is well known that the heat transfer simulations have strict grid requirements on dimensionless distance Y^+ . Therefore, the height of the first layer grid is 0.01 mm and the growth ratio is 1.2 with the maximum aspect ratio equal to 205.08. Those options ensure that the Y^+ value in all numerical cases less than 1 and the proportion of Y^+ values less than 0.91 is up to 99.95%. The detailed information of six sets of meshes is summarized in Table 3.

The effect of Boundary layer number on radial temperature profiles at $z=0.76$ m is shown in Fig. 5. It could be seen clearly that the variation of radial temperature profiles in the Komax is less than 0.26% when $B>13$. Thus, 2537017 grids with $B=13$ in the Komax are selected for the subsequent calculations.

3.2. Governing equations

The three-dimensional steady-state turbulent flow and heat transfer in the tube with Komax and Ross LPD inserts is solved by using ANSYS FLUENT v16.1. The following assumptions are adopted: (1) The fluid is incompressible, (2) The effect of radiation is ignored, (3) The effect of gravity is taken into account,

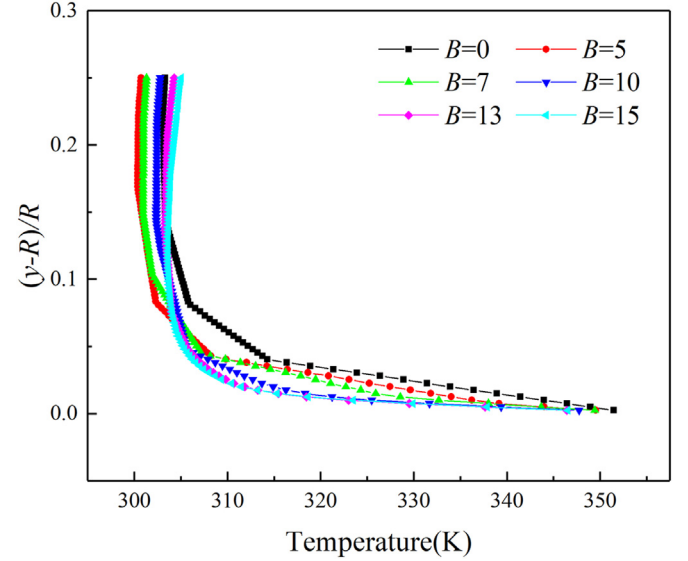


Fig. 5. Grid independence tests of Komax.

(4) The thermophysical properties of water are considered to be temperature-dependent, and the relationship is adopted as follows [43,44]:

$$\rho_w = -3.57 \times 10^{-3} T^2 + 1.88 T + 753.2 \quad (14)$$

$$\mu_w = 2.59 \times 10^{-5} \times 10^{\frac{2383}{T-1432}} \quad (15)$$

$$\lambda_w = -8.35 \times 10^{-6} T^2 + 6.53 \times 10^{-3} T - 0.6 \quad (16)$$

The conservation equations are written as follows [45]:
Continuity equation:

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (17)$$

Momentum equation:

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial(\rho u_i u_j)}{\partial x_j} + \frac{\partial p}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\mu \frac{\partial(u_i)}{\partial x_j} - \rho \overline{u'_i u'_j} \right) + F_g \quad (18)$$

where the source term F_g includes the gravity.

Energy equation:

$$\frac{\partial(\rho T)}{\partial t} + \frac{\partial(\rho u_i T)}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\lambda}{c_p} \frac{\partial T}{\partial x_i} \right) \quad (19)$$

The equations of turbulent kinetic energy (k) and dissipation rate (ω) in SST $k-\omega$ model are given:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left(\Gamma_k \frac{\partial k}{\partial x_j} \right) + G_k - Y_k + S_k \quad (20)$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_j}(\rho \omega u_j) = \frac{\partial}{\partial x_j} \left(\Gamma_\omega \frac{\partial \omega}{\partial x_j} \right) + G_\omega - Y_\omega + D_\omega + S_\omega \quad (21)$$

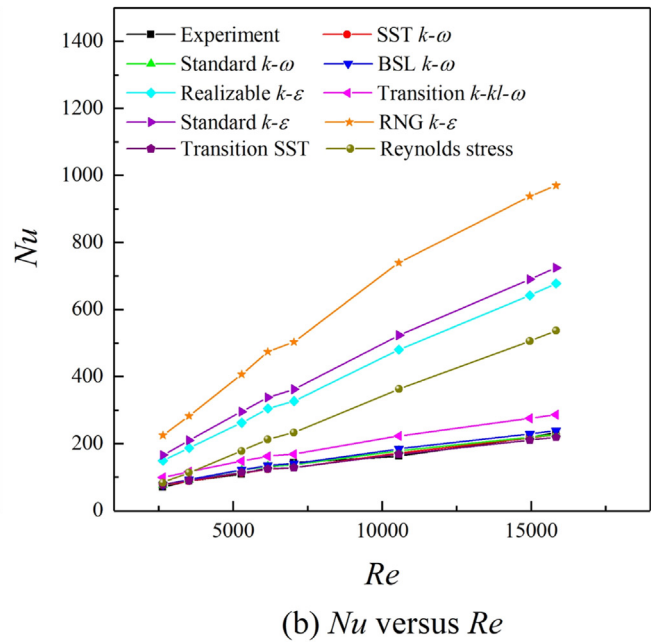
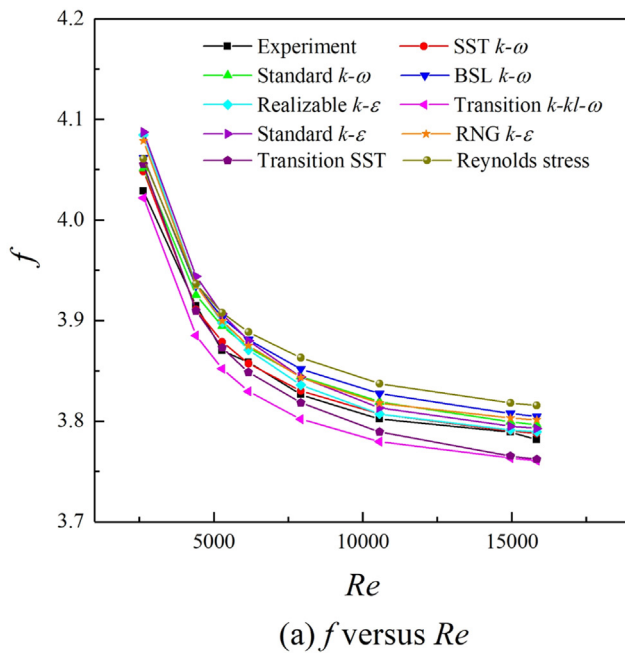


Fig. 6. Model validation based on (a) f and (b) Nu of Komax.

where G_k represents the generation of turbulence kinetic energy, G_ω represents the generation of ω , Γ_k and Γ_ω denote the effective diffusivity of k and ω , Y_k and Y_ω represent the dissipation of k and ω due to turbulence, D_ω represents the cross-diffusion term, S_k and S_ω are user-defined source terms.

3.3. Boundary condition

In the numerical simulation, the inlet is set as velocity boundary condition using User Defined Functions to obtain the fully developed state and reduce the computational cost [31,46,47]. The outlet is set as pressure-outlet. The inlet temperature is set at 293K. No-slip condition is applied on the element walls and tube surface. The uniform heat flux is supplied on the tube surface with the heat flux of $q=50 \text{ kW/m}^2$. In order to make the simulation close to the actual engineering situation, 304 stainless steel is used as the wall materials and fluid-solid-heat coupling is adopted for calculation. The physical property of 304 stainless steel is given as $\rho_s=7.93 \text{ g/cm}^3$, $c_{p,s}=0.5 \text{ kJ/(kg}\cdot\text{K)}$ and $\lambda_s=16.3\text{W/(m}\cdot\text{K)}$, respectively. In steady-state simulation, Pressure-velocity coupling is achieved by using SIMPLEC algorithm. The gradients are computed by the Least Squares Cell-Based discretization method. The spatial discretization of pressure, turbulent kinetic energy terms, specific dissipation rate and energy are obtained by the second-order upwind scheme. First-order upwind discretization is adopted for the momentum. The residual criterion of the energy equation is less than 10^{-8} and those of other equations are set to less than 10^{-6} .

4. Results and discussion

4.1. Verification of numerical model

Under the same boundary conditions, nine different turbulence models are selected to predict the heat transfer and fluid flow in the Komax static mixer. The profiles of Nu and f could be obtained as shown in Fig. 6. It is seen clearly in Fig. 6(a) that f decreases sharply and then declines slowly with the increase of Re . On the other hand, Nu gradually increases with increasing Re .

The errors between numerical results of different models and experimental results are shown in Table 4. It could be seen clearly the minimum errors of Nu and f is 6.91% and 10.21%, respectively. On the other hand, the minimum errors of numerical cases could be simultaneously obtained by the SST $k-\omega$ model. Those indicate that the predicted result from SST $k-\omega$ model is in a good agreement with experimental results. It's worth noting that the over-estimation of Nu predicted by Realizable $k-\epsilon$ and Reynolds stress models is up to 66.19% and 59.95%, respectively. As a result, SST $k-\omega$ model is selected for the following numerical cases. On the basis of the previous research [31], it can be concluded that SST $k-\omega$ model may be recommended for the helical flow simulations because of good robust.

4.2. Fluid flow analysis

Fig. 7 and Fig. 8 show two-dimensional and three-dimensional streamline diagrams of Komax and Ross LPD static mixers, respectively. It can be seen in Fig. 7(a)–7(c) that there is a counterclockwise longitudinal vortex in the tube center, and it becomes smaller and smaller which disappeared before the leading edge of center rectangular plates in the Komax channel. At the same time, four forced vortices generate at the edges of semi-elliptical plates and become stronger and stronger as shown in Fig. 7(d). With the increasing axial position, two semicircular cross-sections are divided by the center rectangular plates. As a result, it could be seen in Fig. 7(e) and Fig. 8(a) that the combination of two forced vortices with opposite swirling direction become a pair of Dean vortices which has smaller scale than the tube diameter and contributes to heat transfer enhancement. The Dean Vortex is gradually weakened and a new longitudinal vortex is gradually merged before the ending edge of the center rectangular plates. A clockwise longitudinal vortex is generated at the transition of the adjacent mixing elements with the opposite helical direction as illustrated in Fig. 7(f) and a similar evolution could be observed from Fig. 7(f) to 7(j).

From the subplots of two-dimensional streamline in the Ross LPD shown in Fig. 7(k) to 7(t), the fluid can be periodically cut and

Table 4
Error between predicted results of different models and experimental data in the Komax.

Model	SST $k-\omega$	Standard $k-\omega$	BSL $k-\omega$	Realizable $k-\epsilon$	Transition $k-k\ell-\omega$	Standard $k-\epsilon$	RNG $k-\epsilon$	Transition SST	Reynolds stress
Error of f	6.91%	8.30%	8.49%	8.02%	7.36%	8.16%	8.28%	7.54%	8.73%
Error of Nu	10.21%	12.82%	13.89%	66.19%	29.77%	69.78%	78.03%	11.14%	59.95%

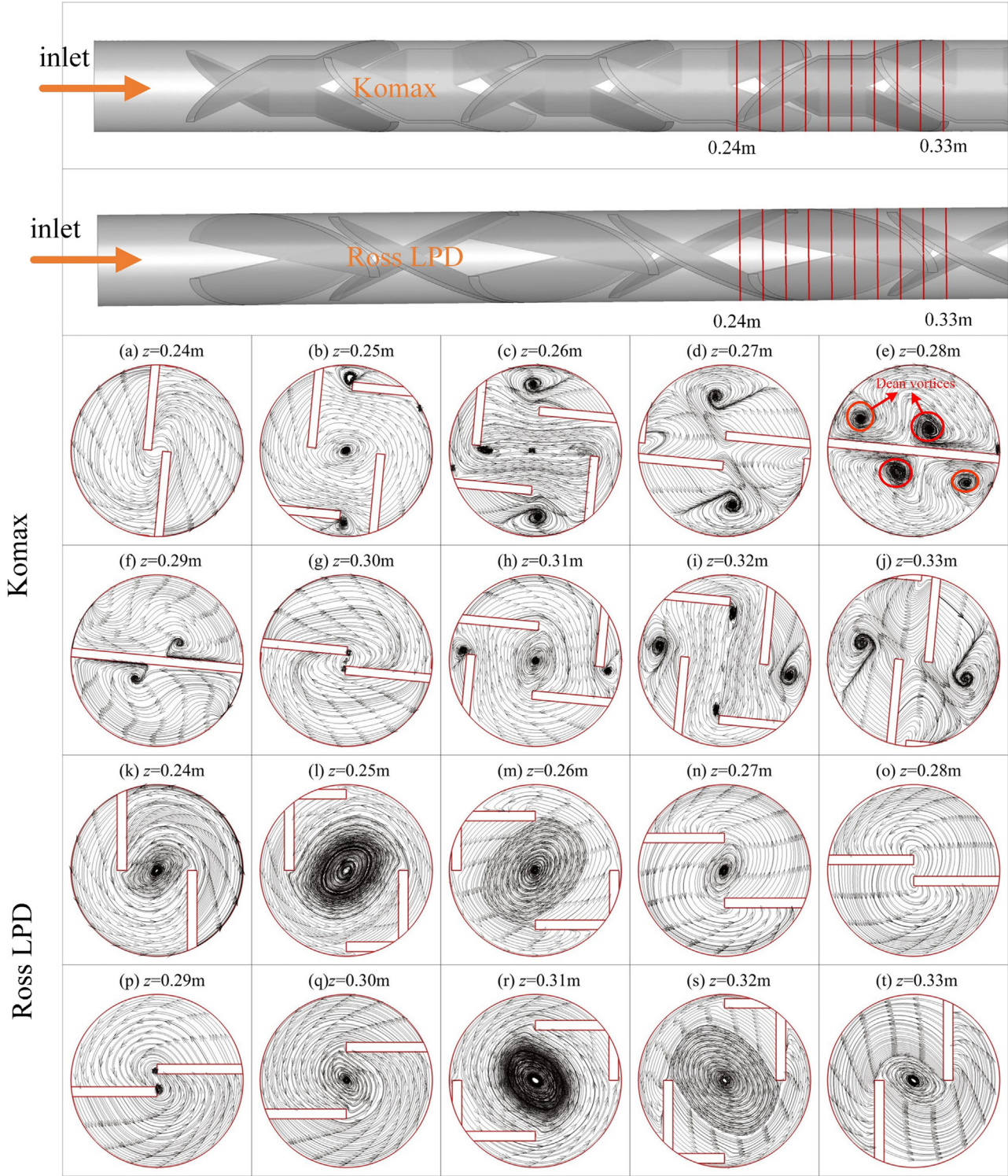


Fig. 7. The streamlines of different cross sections at $Re=10560$ in Komax and Ross LPD.

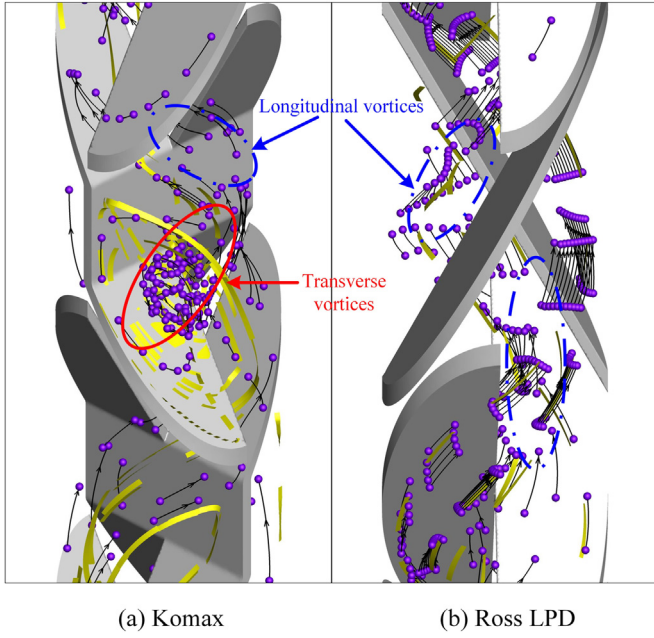


Fig. 8. Three-dimensional streamlines at $Re=10560$ in Komax and Ross LPD.

divided by the downstream element and a few vortices are generated and restricted by the interaction between the tube wall and Ross inserts. It could be seen in Fig. 7(k) to 7(t) and Fig. 8(b) that there are only large counterclockwise longitudinal vortices with the same order of magnitude as the tube diameter. The three-dimensional streamlines diagram illustrates that the transverse vortices perpendicular to the flow direction is formed at Fig. 7(e) and Fig. 8(a) by the secondary flow between the plate and the tube wall.

In order to analyze transverse and longitudinal vortices, some dimensionless numbers are introduced. The secondary flow can be described by the mean absolute value of the vorticity on the nor-

mal cross-section as shown in Eq. (22) [48,49]:

$$J_z = \frac{1}{A} \int_A |\omega_z| dA \quad (22)$$

where A is the cross-section area, and the subscript z indicates the normal direction of the cross-section.

The dimensionless number Se is adopted to evaluate the intensity of secondary flow, and the formula is shown in Eq. (23):

$$Se = \frac{\rho_w (D J_z) D}{\mu_w} = \frac{\rho_w D^2}{\mu_w} J_z \quad (23)$$

The swirl number (S) is a dimensionless number which can reflect the characteristics of swirl intensity more accurately and comprehensively. The swirl number is generally expressed by Eq. (24) [50,51]:

$$S = \frac{\int_0^R u_z u_\theta r^2 dr}{R \int_0^R u_z^2 r dr} \quad (24)$$

where u_z is the axial velocity and u_θ is tangential velocity.

The axial profiles of Se at different Re in Komax and Ross LPD is shown in Fig. 9. Because Ross LPD has a cross blade structure which could make a strong separation effect for the fluid, the secondary flow intensity is high in the zone where the first element overlaps with the front of the second one. However, the intensity of secondary flow gradually decreases in the following because the channel with Ross LPD is easy to form a relatively stable helicoidal flow. In contrast, the complicated structure of Komax prevents the formation of complete helicoidal flow and leads to higher secondary flow intensity. It is seen that the intensity of secondary flow in the Komax is higher than that in Ross LPD. The mean values of Se in the Komax increase from 59.47 to 586.99 and those in the Ross LPD increase from 49.73 to 482.65 with increasing Re . The secondary flow intensity of Komax is 1.19–1.21 times that of Ross LPD under $Re=2640$ – 6160 and the ratio is fixed at 1.22 for $Re \geq 9680$. The results show that the gap of secondary flow intensity between Komax and Ross LPD increases and becomes gradually stable with the increase of Re . The dotted line represents the center of each element, and it could be seen that the maximum values of Se are at the transition of adjacent elements. With the increase of Re , the profiles fluctuates more acutely and the difference

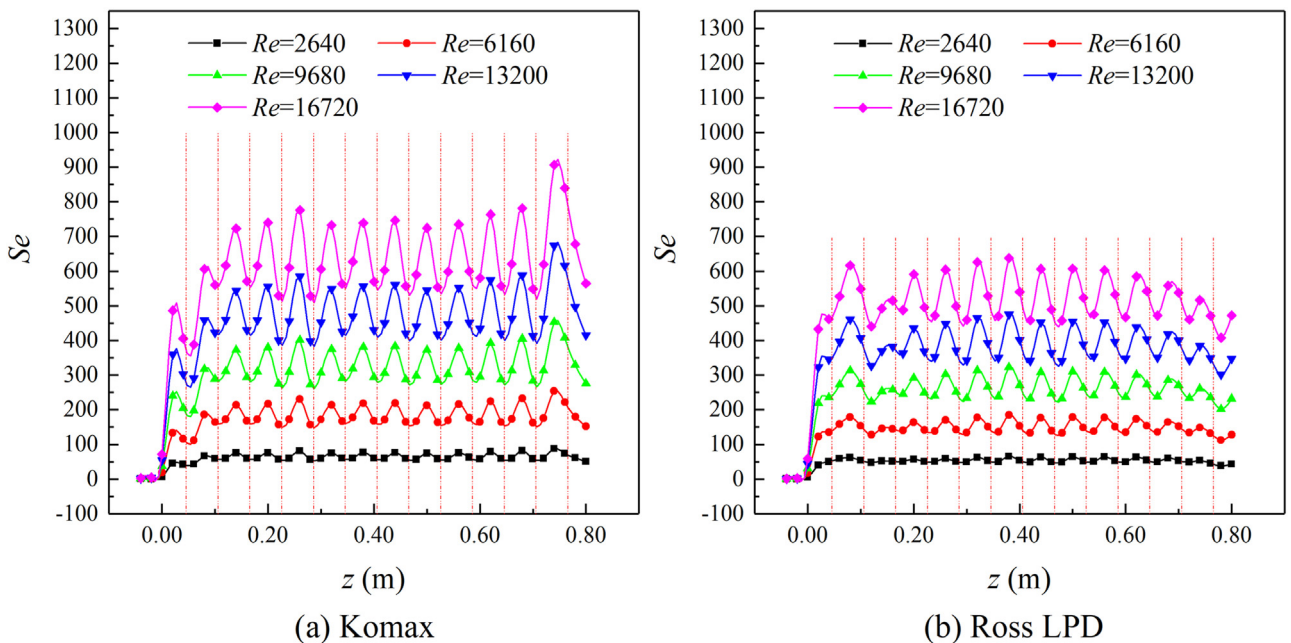


Fig. 9. Axial distributions of Se at different Re in Komax and Ross LPD.

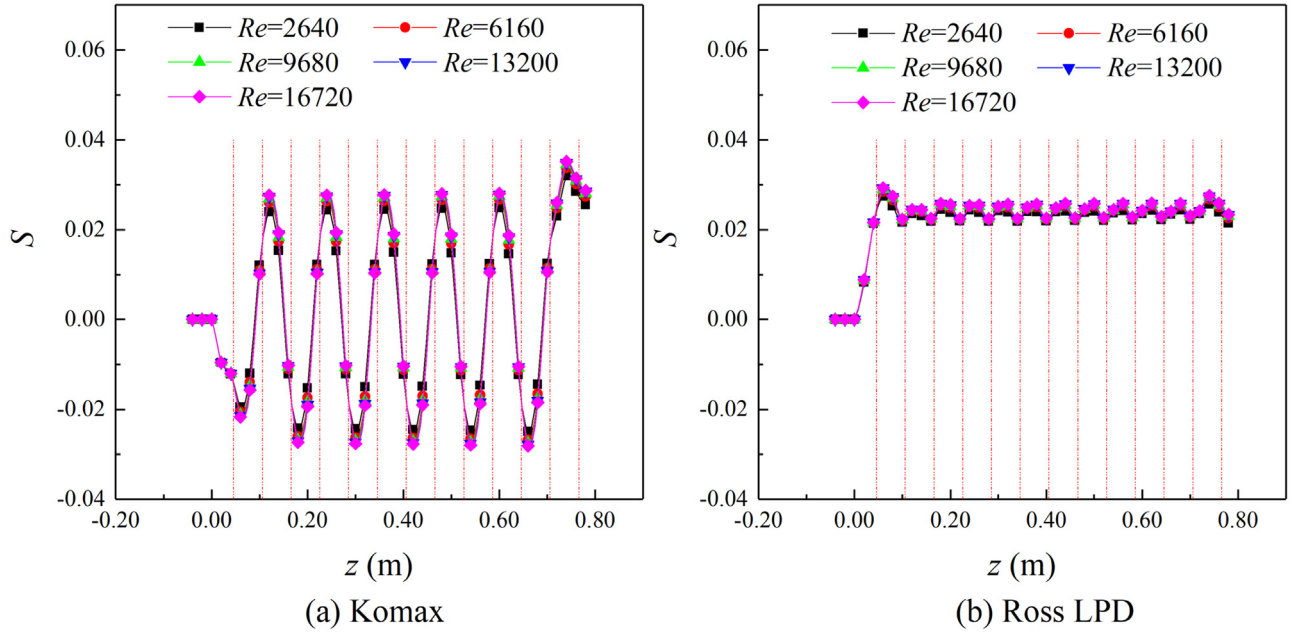


Fig. 10. Axial distributions of S at different Re in Komax and Ross LPD.

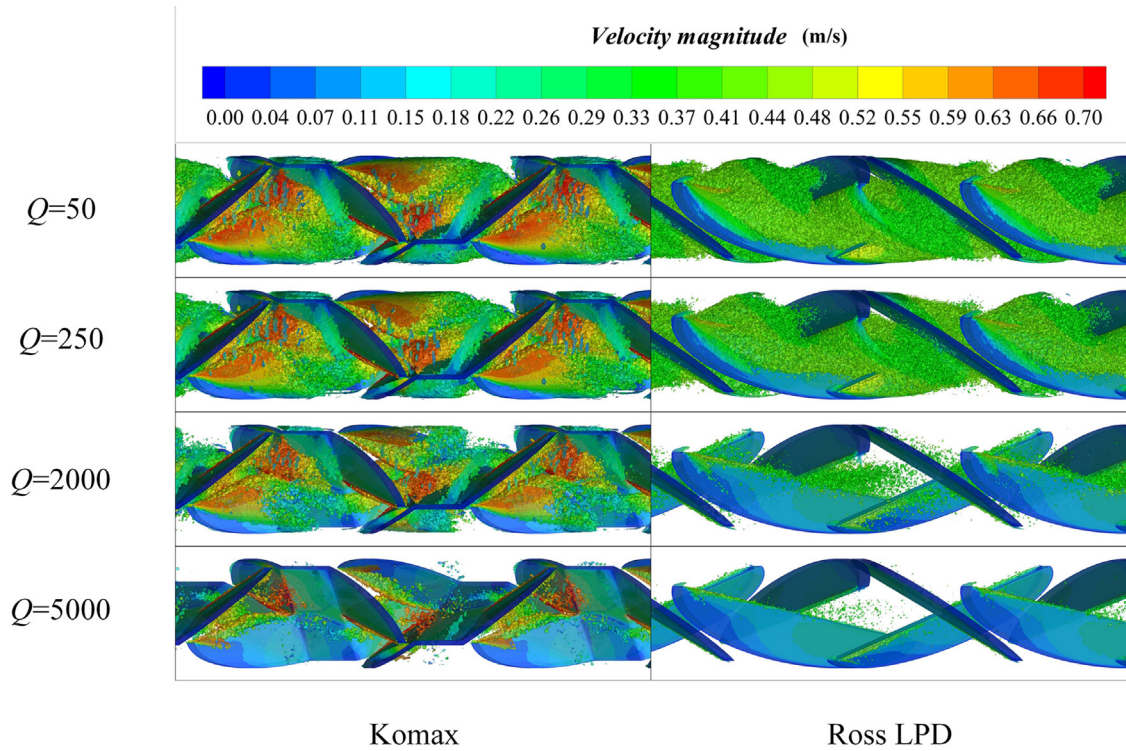


Fig. 11. The iso-surface overlap between Q criterion and velocity magnitude contour for $Re=10560$.

between Komax and Ross LPD becomes much larger. The ranges of relative standard deviation (RSD) of Se are 18.88–187.68 and 14.67–140.54 for Komax and Ross LPD, respectively.

Fig. 10 depicts the axial profiles of S at different cross-sectional positions of Komax and Ross LPD. Due to the opposite Komax inserts, the helical directions of longitudinal vortex are periodical counterclockwise and clockwise which induced the profiles of S varies with sinusoidal function. It can be seen clearly that the S values are often insensitive to Re . There are 12 peaks in the pro-

files of S and these peaks are located at two thirds length of the elements. The absolute ranges of local maximum values of S in the Komax and Ross LPD are 0.032–0.035 and 0.027–0.029, respectively. However, the fluid in Ross LPD rapidly forms a relatively stable helicoidally flow after encountering the elements and the tangential velocity does not change along the axial direction. As a result, S of the cross-section does not fluctuate greatly.

It is well known that Q criterion is used to identify the multi-scale vortices in the flow field [52]. Fig. 11 represents the super-

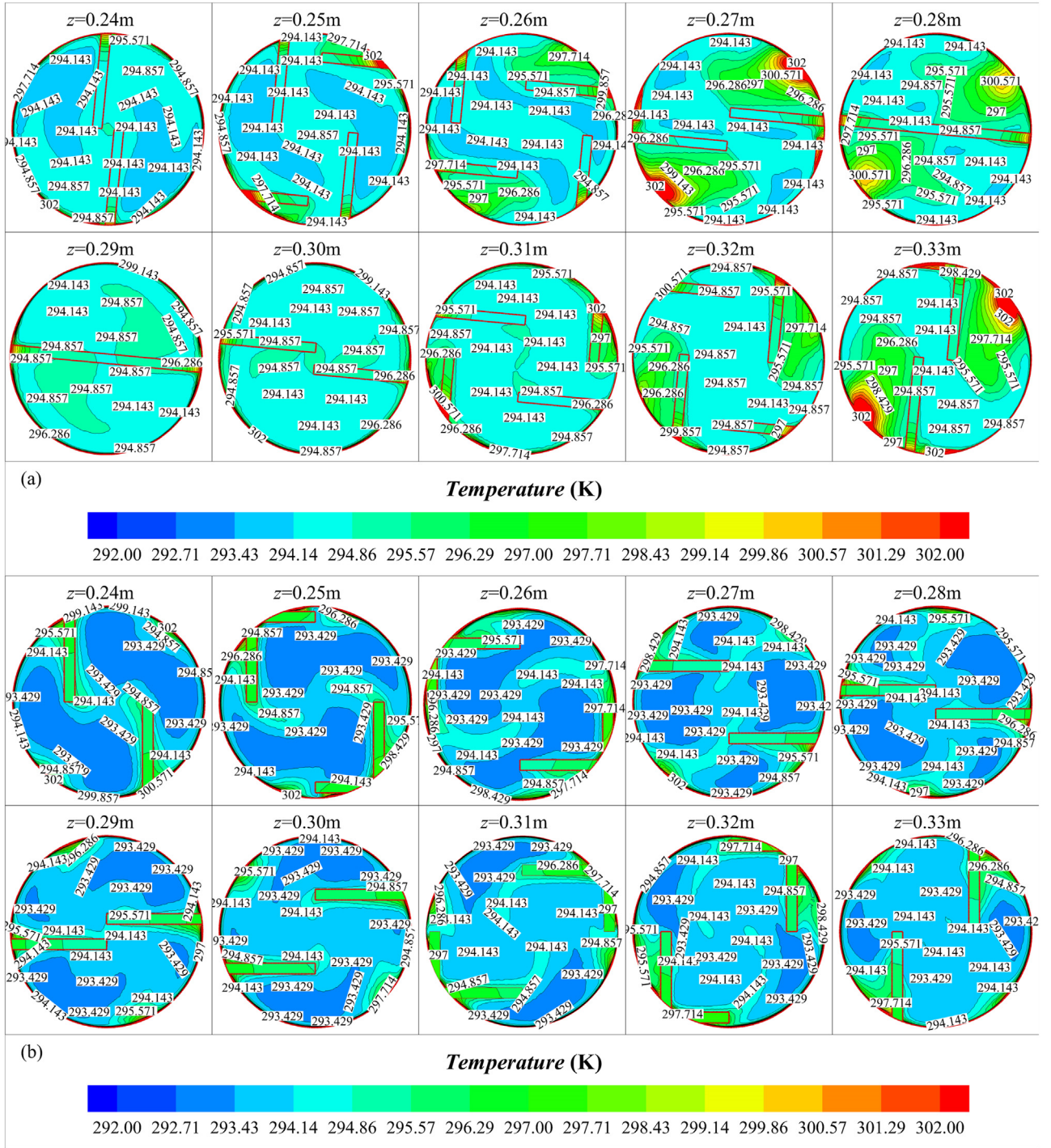


Fig. 12. The cross-sectional temperature contour at $Re=10560$ in (a) Komax and (b) Ross LPD.

imposed iso-surface maps of Q value and velocity contour in the Komax and Ross LPD, respectively. From the subplots at $Q=2000$ and 5000 , it can be found that Komax has larger vortices region in the flow field and more obvious vortices structure near the blade edges compared with the Ross LPD. From the graph subplots at $Q=50$ to 5000 , it can be found that the maximum velocity magnitude of Komax is more than 0.70 m/s. However, the maximum of velocity magnitude is just up to 0.59 m/s in Ross LPD. The match between higher velocity magnitude and larger vortices in-

dicates that the Komax insert has a higher efficiency in adjusting the velocity fields. Moreover, a narrow flow channel is formed in the overlapping zone of the two elements, which compresses the fluid and make the fluid disturbance be stronger in a jet-like manner.

4.3. Heat transfer analysis

Fig. 12 shows the cross-sectional temperature distributions in Komax and Ross LPD at $Re=10560$. From the temperature contour

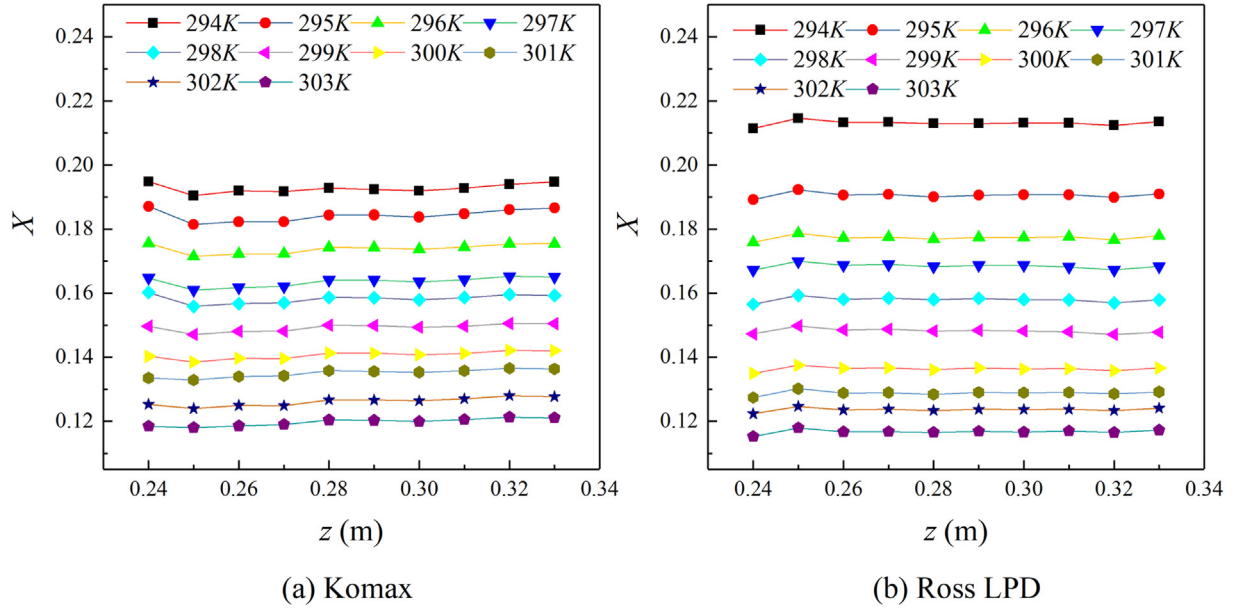


Fig. 13. The area ratio of temperature range.

at $z=0.27$ m, the temperature values of bulk flow in the Komax and Ross LPD flow field range from 295.57 K to 302.00 K and 292.00 K to 294.14 K, respectively. From the temperature contours of $z=0.29$ m–0.31 m, the temperature ranges in the core flow of the Komax and Ross LPD are 294.14 K–294.86 K and 292.71 K–294.14 K, respectively. It is obvious that the internal temperature of Komax is higher than that of Ross LPD and the temperature in the core of the flow field become more and more uniform. It accords with the core flow theory [53] that the temperature of central flow is uniform and the temperature gradient of boundary layer is large. It could be seen from the temperature contours at $z=0.27$ m and $z=0.33$ m that two symmetric high temperature regions appear near the wall. Combined with Fig. 7, it can be found that the high-temperature zones are generated by the interaction between small vortices and large vortices which increases the residence time of the fluid and the local heat absorption. However, there are few irregular disturbances in Ross LPD, which leads to slow fluid mixing and heat transfer.

Fig. 13 demonstrate the ratio of area with different temperature range for Komax and Ross LPD, respectively. The cross-sectional average temperature has a slight increase and is stable with increasing length of flow passage. The X values of low temperature in Ross LPD is significantly larger than that in Komax. The values of X in Ross LPD decrease from 1.10 to 1.02 times that of Komax for $T=294\text{K}$ – 297K . There is a turning point at $T=298\text{K}$ and the X with higher temperature of Komax is larger than that in the Ross LPD.

To evaluate the comprehensive heat transfer performance, PEC is defined as follows [54,55]:

$$PEC = \left(\frac{Nu}{Nu_0} \right) \left(\frac{f}{f_0} \right)^{-1/3} \quad (25)$$

where Nu_0 and f_0 are the Nusselt number and friction factor of a smooth tube, respectively.

As plotted in Fig. 14(a), f slightly decreases and Nu increases with the increasing Re in Komax and Ross LPD. The ranges of f are 3.78–4.05 and 0.69–1.13 in Komax and Ross LPD, respectively.

The Nu ranges from 67.66–248.20 and 46.10–182.26 for Komax and Ross LPD, respectively. Fig. 14(b) shows the variation of the ratio f/f_0 and Nu/Nu_0 with Re . The ratio f/f_0 ranges from 96.62–102.66 and 367.31–551.60 for Ross LPD and Komax, respectively. Obviously, the ratio f/f_0 increases significantly with increasing Re for Komax, but that of Ross LPD is more stable. The ratio Nu/Nu_0 of Komax and Ross LPD is in the range of 9.10–16.75 and 6.08–11.42, respectively. Komax could obviously enhance the heat transfer coefficient by 46.7%–49.6% at the cost of Darcy friction factor increasing 2.80–4.37 times. For Ross LPD, f and f/f_0 do not increase significantly with the increase of Re for the longitudinal vortices with the same scale as the flow field are formed in the flow field. The internal disturbance of the flow field is relatively simple, so it could not lead to better convection heat transfer enhancement. Apart from the longitudinal vortices with the same flow field scale, there are transverse vortices smaller than the flow field scale in Komax due to the interaction between fluid and elements resulting in a significant increase of f/f_0 with increasing Re . The interaction between transverse and longitudinal vortices improves the convective heat transfer capacity of the fluid, so Nu and Nu/Nu_0 increase sharply with the increasing Re . Fig. 14(c) shows the overall heat transfer performance with increasing Re . It is obvious that the PEC has the best performance in lower Re range while it decreases rapidly in higher Re range. Moreover, the variations of PEC values in the higher Re are small which are still above 1. In addition, the PEC values of Ross LPD are significantly larger due to its simple structure, less flow field disturbance and lower flow resistance. The ratios for PEC in Ross LPD to that in Komax is less than 1.1 when Re is smaller than 5280, which indicates that there is little difference between two static mixers. The superiority in PEC is highlighted for Ross LPD when Re is greater than 5280 and the ratio exceeds 1.3 at $Re=17600$.

In order to further reveal the heat transfer mechanism of Komax and Ross LPD, the local Nusselt number Nu_{local} is evaluated under different Re as shown in Fig. 15. Fig. 15(a) shows the 3-D Nu_{local} around a tube fitted with Komax static mixer. Obviously, the regions with higher Nu_{local} is near the interfaces between tube wall and center rectangular plates of Komax. This is because the generation of the transverse vortexes described in Fig. 8 weakens

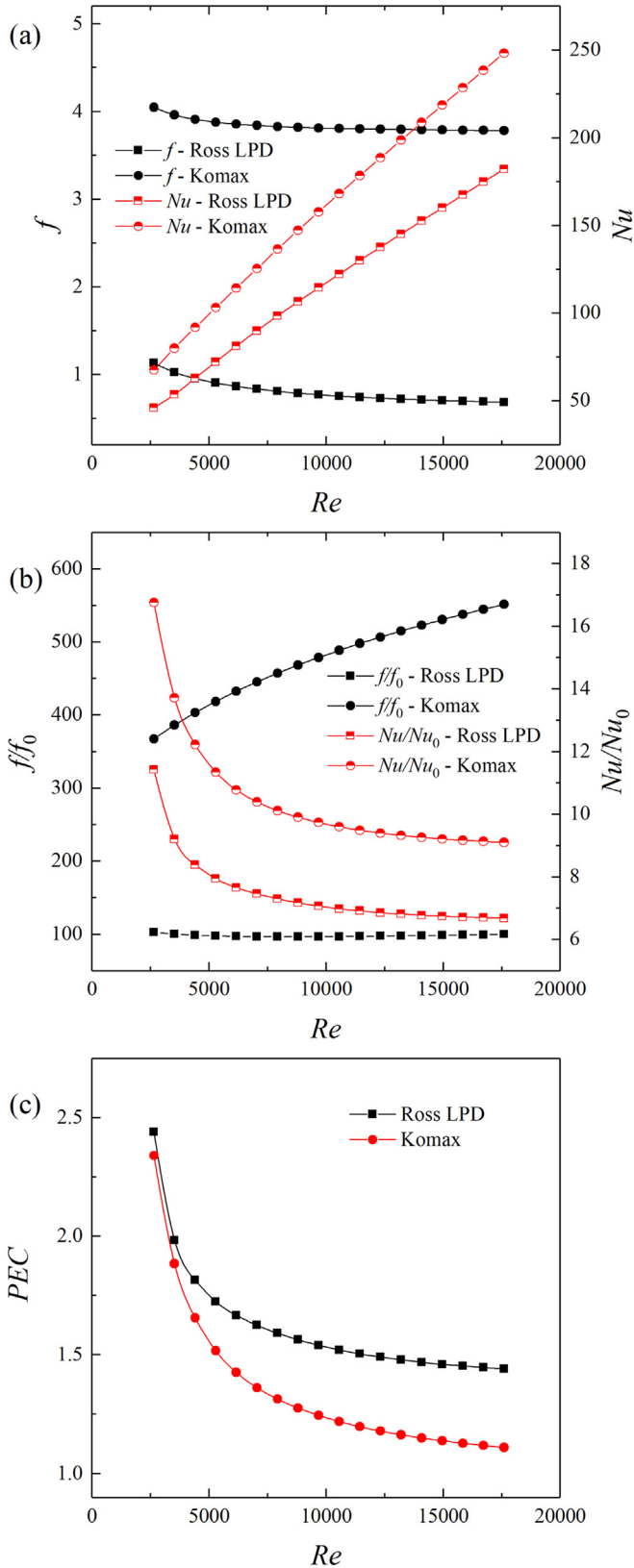


Fig. 14. The comparison of heat transfer and flow resistance.

the thickness of boundary layer and thus enhances the heat exchange between the wall and the fluid. It can be seen in Fig. 15(b) that the cross-sectional Nu_{local} of Komax decreases first and then increases with the increase of axial position under different Re . The turning point is early generated with the increase of Re . For $Re=5280-14960$, the turning points of Nu_{local} are located from the tenth to the fifth element. It can be found in Fig. 15(c) that Nu_{local} of Ross LPD decreases with the increase of axial position under different Re . It is because of intensity attenuation of longitudinal vortices on boundary layer disturbance in Ross LPD with the axial location. Moreover, the intensity of transverse vortices is higher than that of longitudinal vortices with the increase of Re in Komax, which may lead to the early generation of turning point of Nu_{local} .

Guo et al. [56] first proposed the concept of field synergy at the end of the last century and introduced it in the field of convective heat transfer. This theory explored the principles and methods to improve the effect of convective heat transfer from the perspective of different physical fields and interactions. Tao et al. [57–60] numerically and experimentally verified the principle of field synergy in the process of convective heat transfer. Based on the idea of field synergy, Liu et al. [61] studied the heat exchange in the laminar and proposed that each point in the flow field can be analyzed by using field synergy theory. Zhou et al. [62] proposed five methods to evaluate the synergy angle based on the field synergy theory.

The volume-weighted average synergy angle between the velocity and the pressure gradient is θ , the volume-weighted average synergy angle between the velocity and the temperature gradient is β . The θ and β can be expressed as follow:

$$\theta = \frac{\sum \arccos \frac{\vec{u} \cdot \nabla p}{|\vec{u}| |\nabla p|} \cdot V_i}{\sum V_i} \quad (26)$$

$$\beta = \frac{\sum \arccos \frac{\vec{u} \cdot \nabla T}{|\vec{u}| |\nabla T|} \cdot V_i}{\sum V_i} \quad (27)$$

Fig. 16 depicts the variations of θ and β with Re in Komax and Ross LPD. The variations of θ are from 61.7° to 63.3° and 71.2° to 73.4° for Komax and Ross LPD, respectively. The θ is commonly used to represent the pump power consumed by fluid. Generally, it is believed that the pump power consumption increases with increasing θ . The θ in Komax is significantly lower than that of Ross LPD which shows that the overall pump power consumption of Komax is lower than that of Ross LPD. The variation of β are from 83.4° to 84.3° and 86.0° to 85.1° for Komax and Ross LPD, respectively. The β reflects the synergistic degree of heat transfer. The smaller β is, the velocity of the fluid has the same direction with heat diffuse. That is, the stronger convective heat transfer is between the fluid and the wall. The β of the Komax is lower than that of the Ross LPD due to the interaction between transverse and longitudinal vortices in the flow field thus enhancing convective heat transfer. Therefore, it can be concluded that the fluid interaction between large longitudinal and small transverse vortices not only can reduce the power consumption but also enhance convective heat transfer between fluid and wall surface.

According to the experimental data of each variable, the related expressions of f and Nu for Komax and Ross LPD in the range of $Re=2640-17600$ are established respectively, which can be expressed as follows:

Komax:

$$Nu = a_1 \times Re^{a_2} \times Pr + a_3 \quad (2640 \leq Re \leq 17600) \quad (28)$$

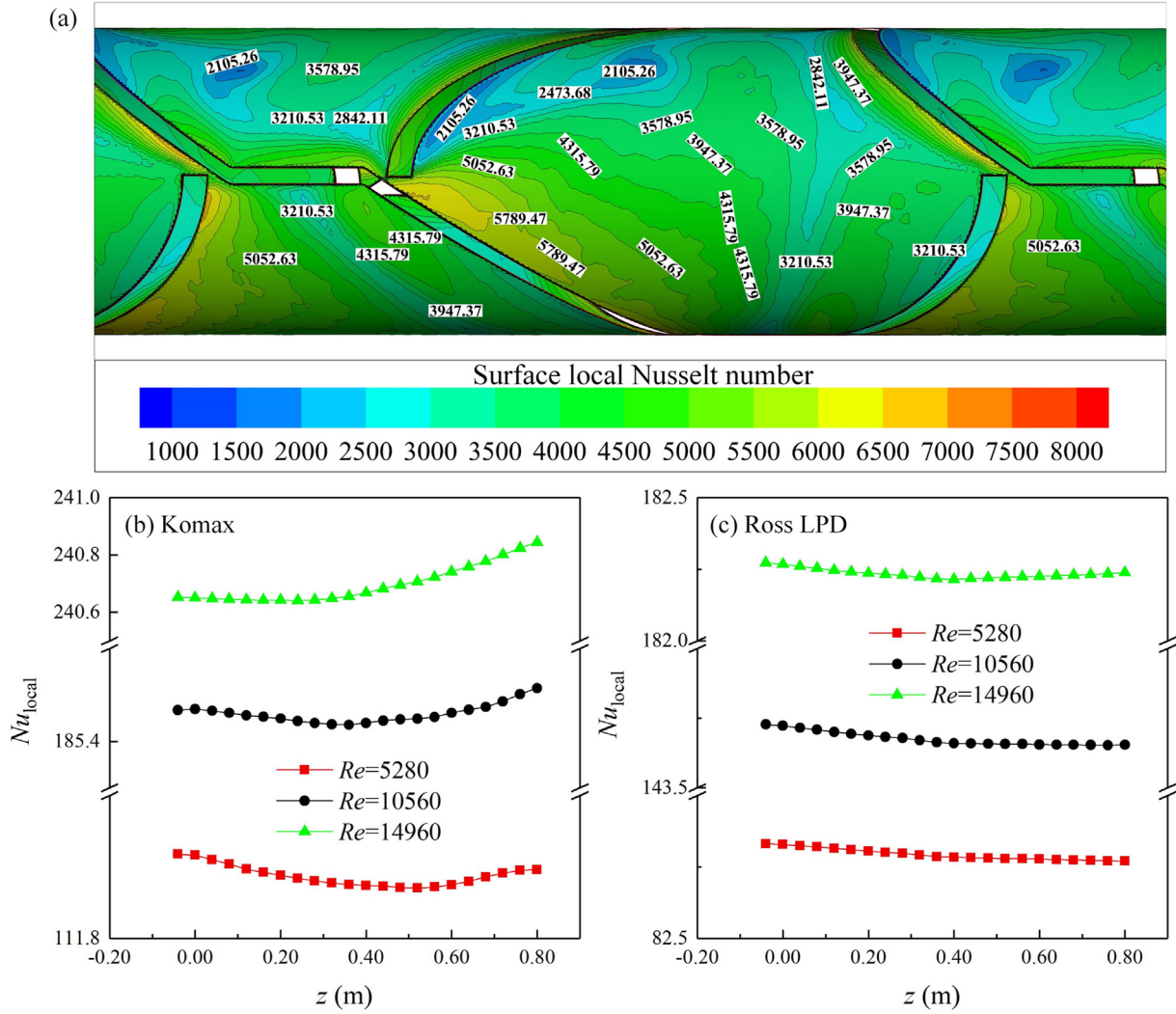


Fig. 15. The distributions of Nu_{local} : (a) The 3-D Nu_{local} around a tube fitted with Komax static mixer, (b) the Nu_{local} of Komax and (c) Nu_{local} of Ross LPD versus axial positions.

where $a_1 = 0.0844$, $a_2 = 0.6506$, $a_3 = -23.3013$.

$$f = b_1 \times Re^{b_2} + b_3 \quad (2640 \leq Re \leq 17600) \quad (29)$$

where $b_1 = 1.0458 \times 10^3$, $b_2 = -0.9162$ and $b_3 = 3.4666$.
Ross LPD:

$$Nu = c_1 \times Re^{c_2} \times Pr + c_3 \quad (2640 \leq Re \leq 17600) \quad (30)$$

where $c_1 = 0.0077$, $c_2 = 0.8443$, $c_3 = 16.7146$.

$$f = d_1 \times Re^{d_2} + d_3 \quad (2640 \leq Re \leq 17600) \quad (31)$$

where $d_1 = -0.5842$, $d_2 = 0.1328$ and $d_3 = 2.7706$.

As shown in Fig. 17, the predicted values of f and Nu derived from the correlations are compared with the experimental values. For the Komax, the errors of Nu and f are less than 5.13% and 4.22%, respectively. For Ross LPD, the errors of Nu and f are less than 5.96% and 5.87%, respectively. The results show that the predicted empirical correlations for both static mixers are in good agreements with the experimental values.

4.4. Energy dissipation and mixing

Fig. 18 shows the radial distribution of turbulent kinetic energy (TKE/ k) and turbulent dissipation rate (ε) in Komax and Ross LPD at cross section of $z=0.27$ m. Mesoscopic mixing and microscopic mixing can be respectively characterized by TKE and ε [63]. As shown in Fig. 18(a) and 18(c), the TKE of Komax was significantly higher than that of Ross LPD, and it was obviously higher in the center and lower on both sides. This indicates that the mesoscopic mixing ability of Komax is much higher than that of Ross LPD, and it increases 21.05, 13.98 and 13.21 times under $Re = 5280$, 10600 and 14960, respectively. The ε in Fig. 18(b) and 18(d) has the same variation trend as TKE. The ε in Komax is much higher than that in Ross LPD which increases 87.71, 53.17 and 49.92 times at $Re=5280$, 10600 and 14960, respectively. Mixing performance of Komax is better than that of Ross LPD at the micro level. From the perspective of mixing, the effect of single longitudinal vortices is obviously not as good as that of interaction by longitudinal and transverse vortices with multiple scales.

The power spectrum P_w of outlet instantaneous pressure fluctuation series at $Re=10560$ of Komax and Ross LPD is depicted as

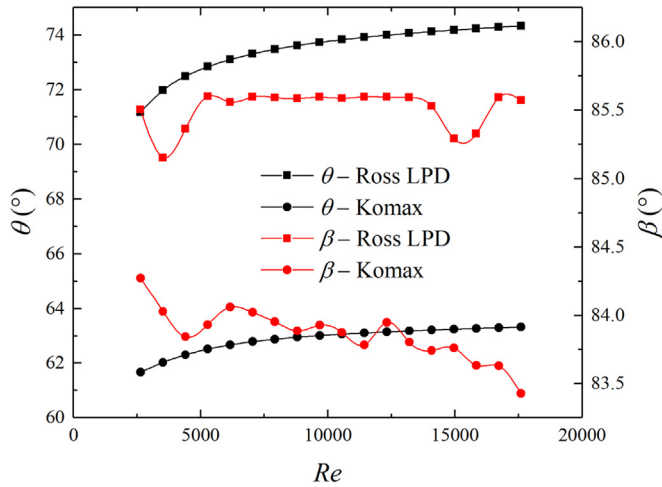


Fig. 16. The volume-weighted average synergy angle varies at $Re=10560$.

shown in Fig. 19(a) and 19(b). From Fourier transform of pressure pulsation signals at outlet section, the first-order primary frequency of turbulent flow are 0.33 Hz and 0.13 Hz, respectively. And the ranges of first-order primary frequency is 0.19 Hz–0.69 Hz and 0.05 Hz–0.18 Hz for Komax and Ross LPD with increasing Re , respectively. There are some other peaks besides the main peak which indicates that the pressure pulsation signals have obvious chaotic characteristics. The power spectrum distributions of pressure pulsation signals depict the turbulent characteristics of multi-scale vortices. Low frequency pulsation is induced large-scale vortices and the small-scale vortices determined by viscous force has high-frequency pulsation.

As shown in Fig. 19(c) and 19(d), the scale relationship between power spectrum and frequency satisfies power function attenuation in logarithmic coordinates. In the range of $Re=2640$ –17600, the logarithmic power spectrum is fitted by selecting appropriate scale region. The scale ϕ value varies with Re , and it fluctuates between 0.97–1.99 and 1.12–2.91 for the Komax and Ross LPD, respectively. And ϕ has turning points at $Re=7040$ and 6160 for Komax and Ross LPD which indicates that both pressure fluctuations

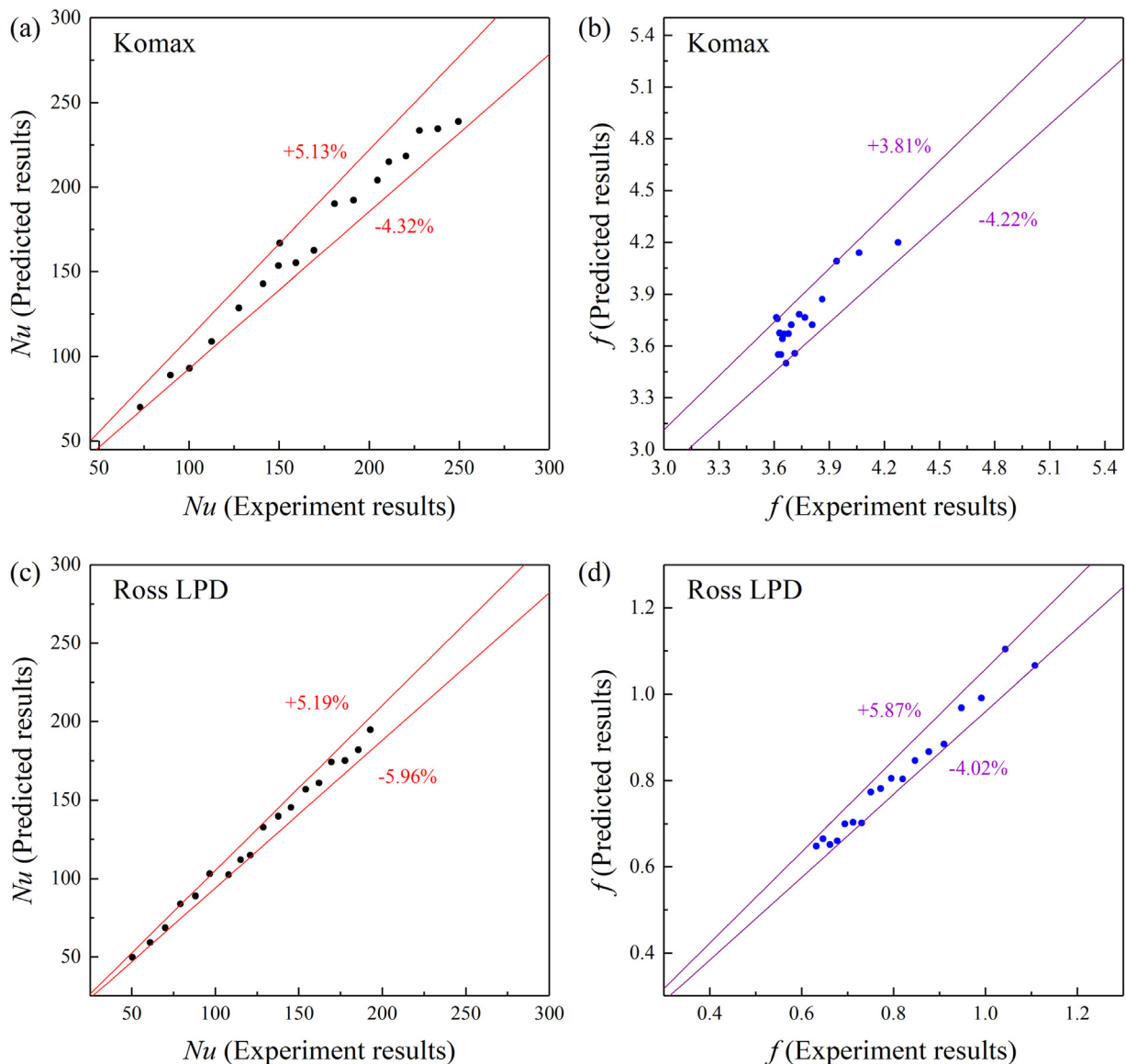


Fig. 17. Predicted data of Nu and f versus experiment data for Komax and Ross LPD.

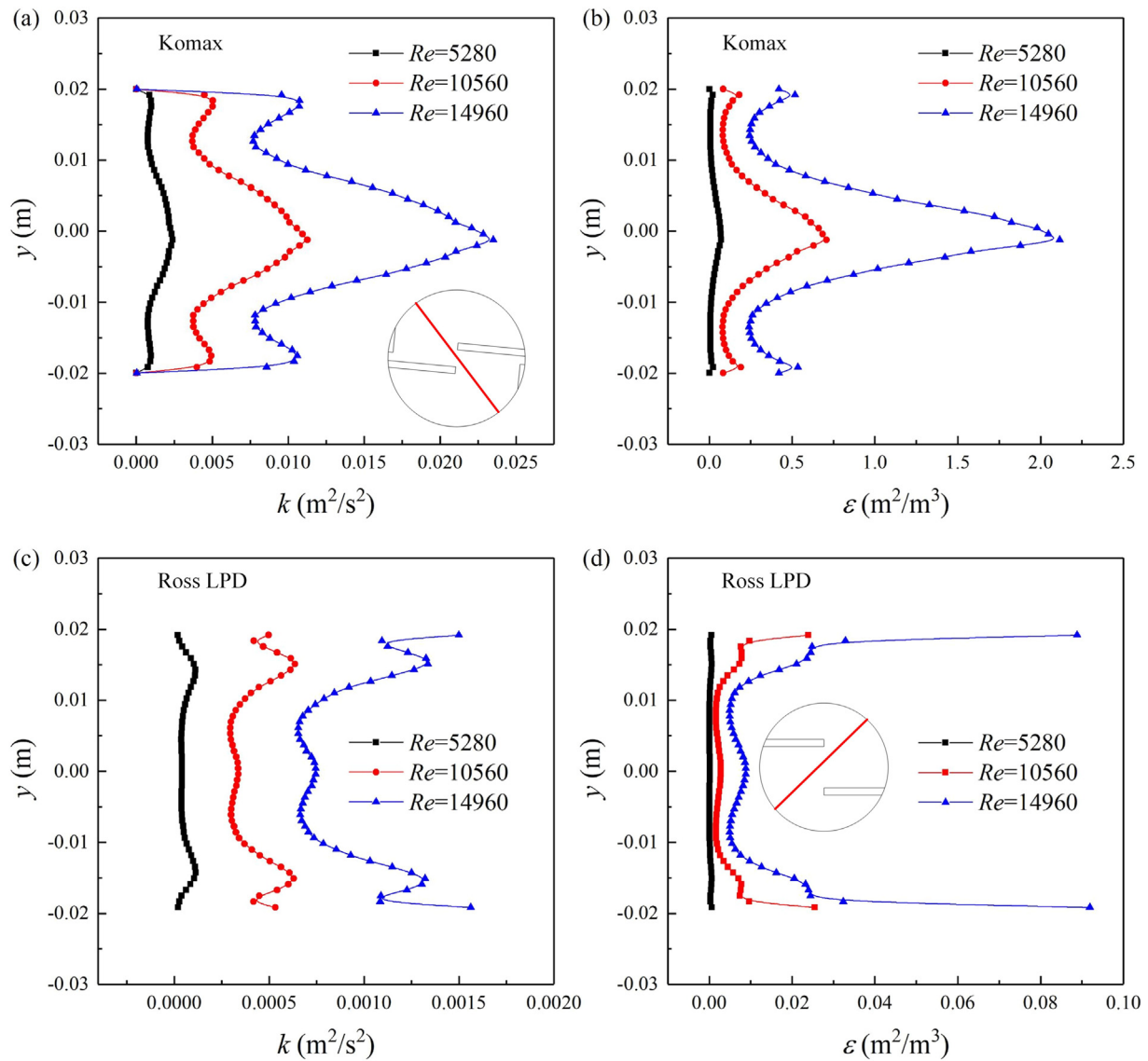


Fig. 18. Evolution of turbulent kinetic energy and turbulent dissipation rate for Komax and Ross LPD.

have fractal Brownian motion characteristics. Furthermore, Komax plays a positive role in improving the capacity of heat transfer and mixing by generating vortices with scale smaller than the flow field.

4.5. Comparison with previous work

The PEC of the tube with Komax and Ross LPD inserts are compared with those of tubes mounted with other inserts, such as decreasing/increasing coil of Eiamsa-ard et al. [8], multiple twisted tapes of Hong et al. [11], tape with trapezoidal-cut of Murugesan et al. [15], twisted tape of Bhattacharyya et al. [17], full-length tape of Eiamsa-ard et al. [20], modified Kenics static mixer of

Meng et al. [27], rectangular winglet vortex generators of Liu et al. [38].

Fig. 20 shows that the PEC values decrease with increasing Re . The PEC of the tube with Komax and Ross LPD inserts are better than that in most of inserts. Komax and Ross LPD have better overall heat transfer performance than modified Kenics static mixer at $Re > 7000$ and $Re > 5000$, respectively. Moreover, Komax and Ross LPD have better overall heat transfer performance than twisted tape at $Re < 4000$ and $Re < 13000$ respectively. Hence, further modifications of element structure gives an important implications for enhancement mechanism of heat and mass transfer in the Komax and Ross LPD static mixers.

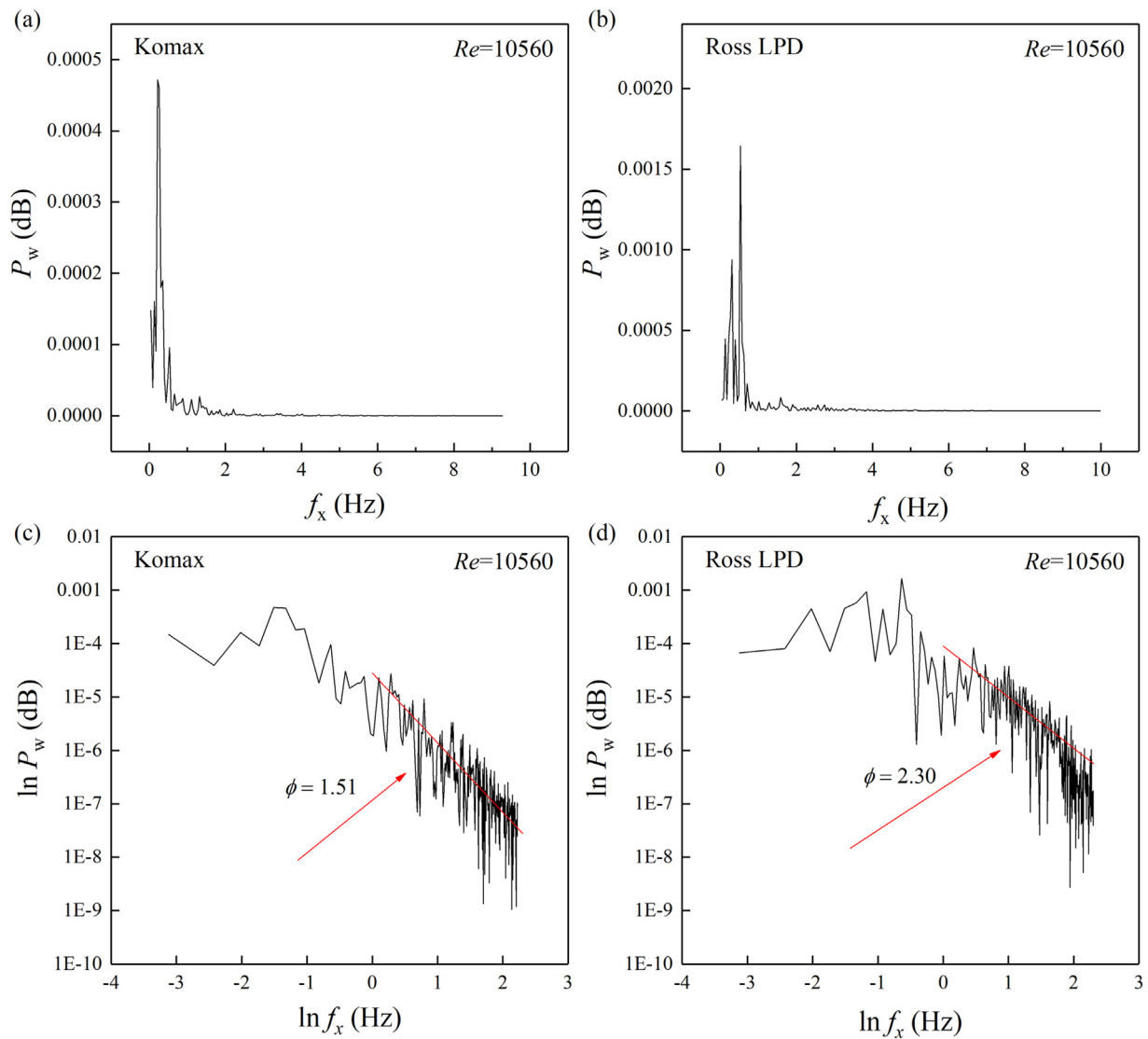


Fig. 19. The power spectrum and scaling exponent of outlet pressure fluctuation time series.

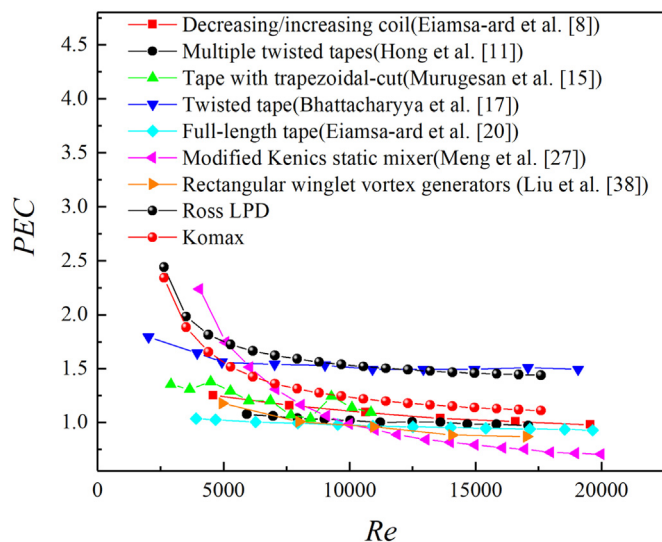


Fig. 20. Comparison of PEC with previous work.

5. Conclusion

The turbulent flow and heat transfer characteristics in the Komax static mixer are investigated experimentally and numerically with uniform heat flux boundary in the range of Re from 2640 to 17600. The enhancement mechanism of Komax inserts has been evaluated based on Ross inserts by the vortex evolution, PEC, field synergy principle and power spectrum analysis, respectively.

The SST $k-\omega$ model could be considered as a recommend choice for the heat and mass transfer simulations in the helical static mixer which is in good agreements with the experimental results. Dean vortices with smaller scale than the longitudinal vortices in the Ross LPD are periodically induced by the Komax. The magnitude of secondary flow intensity and swirl intensity of Komax are larger than that of Ross LPD. Komax could obviously enhance the heat transfer coefficient by 46.7%–49.6% at the cost of Darcy friction factor increasing 2.80–4.37 times. Therefore, Komax insert are recommended at $Re < 5280$ and Ross LPD would be a good alternative due to much lower pressure drop for Re much higher than 5280.

Based on the principle of field synergy, the Komax velocity field and temperature field have good synergy performance, which indicates that the interaction between large-scale longitudinal vortices and small-scale transverse vortices is more conducive to enhancing heat transfer than the single longitudinal vortices. Based on the experimental data, four new empirical correlations based on the experimental data is developed to predict the Nu and f with the errors less than $\pm 6.20\%$.

The frequency domain analyses of pressure fluctuation indicate the dominated frequency of turbulent flow is lower induced by larger scale vortex. There are some other peaks besides the main peak which indicates that the pressure pulsation signals have obvious chaotic characteristics and the chaotical flow could enhance the heat and mass transfer.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge funding support for this research from the Key Scientific Research Project of Education Department of Liaoning Province (LJKZ0429), Shenyang Science and Technology Project (RC200032), National Natural Science Foundation of China (21476142), Distinguished Professor of Liaoning Province (LCH [2018] No. 35), the Science and Technology Research Project of Liaoning BaiQianWan Talents Program (201892151), and Liaoning Natural Science Foundation Project (2019-ZD-0082). We thank the referees for their enlightening remarks which helped us improve the manuscript.

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